

BỘ GIÁO DỤC VÀ ĐÀO TẠO
ĐẠI HỌC KINH TẾ TP. HỒ CHÍ MINH
VIỆN ĐỔI MỚI SÁNG TẠO



USER INTERFACE ANALYSIS REPORT
COURSE: Human Computer Interaction

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GROUP 4

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PERSONAL REFLECTION

Throughout the process of participating in the Library+ website design project, I had the opportunity to observe, discuss, and directly contribute to various stages — from identifying user needs to building the prototype and conducting testing. Looking back on the entire journey, I began to reflect more deeply on the design aspects that our team implemented, as well as the HCI-related issues that may have been overlooked or still have limitations.

Below are some of my personal reflections on the project, with a particular focus on other potential HCI problems, alternative design choices, and how I might do things differently if I had the chance to start over.

1. HCI Issues in the Product

Seat Map Interface Contains Too Much Detail

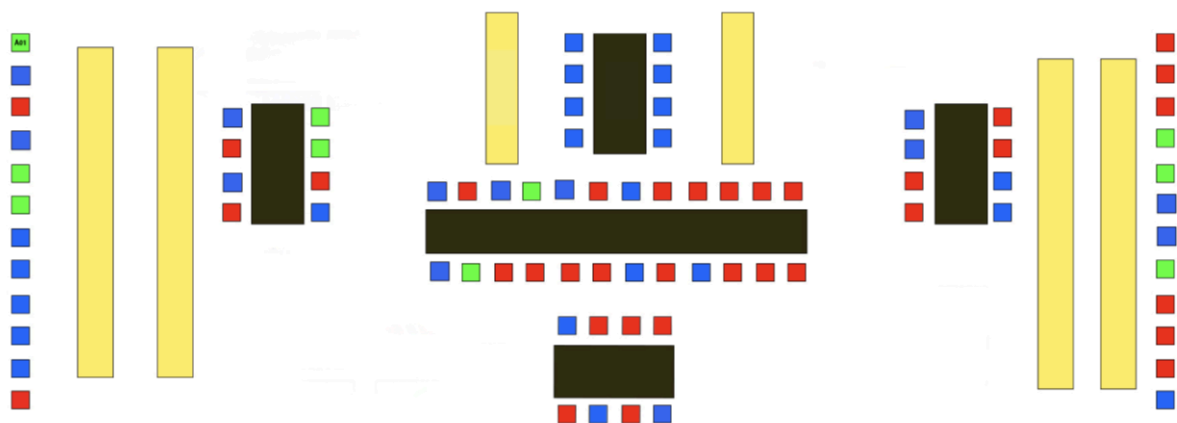


Figure 1. Seating Diagram

The current seat map interface in the prototype presents a large amount of information at once, including: specific seat numbers for each position, color indicators showing seat status (available, reserved, or selected), and a densely packed seating layout. Displaying so many visual elements on a small screen increases cognitive load for users, especially for those interacting with the system for the first time.

This not only makes the interface visually overwhelming but also causes users to spend more time identifying the information they need (such as available seats or preferred locations). The constant visual processing required to differentiate colors and locate specific seats makes the seat selection process less efficient, ultimately reducing the overall user experience.

According to HCI (Human-Computer Interaction) principles, an effective interface should ensure simplicity, visibility, and minimize users' cognitive load. However, the current interface violates these principles by displaying dense information without appropriate layering or visual support.

Lack of Spatial Context (e.g., Main Door, Windows, Aisles)

In the current seat map, the interface focuses solely on functional objects like seats, desks, and bookshelves, while omitting important spatial context elements such as the main entrance, windows, aisles, entry/exit points, or designated areas (e.g., quiet zones, group work areas). The absence of these spatial landmarks makes it difficult for users to mentally visualize the actual location of a seat in the room, which impacts their decision-making ability.

In HCI, providing adequate spatial context aligns with the principle of “cognitive mapping” — helping users form a clear mental image of the physical environment. Without orientation cues like entrances or walkways, users may struggle to determine the front or back of the room, convenient spots, or accessible locations. This reduces the intuitiveness and effectiveness of the interface, especially for those unfamiliar with the library or the layout.

Moreover, the lack of comprehensive spatial representation reduces the realism of the prototype, making it less capable of simulating the real-life experience. Users might select a seat only to later discover it's near a noisy area, far from the door, or inconvenient to access.

Legends for Seat, Study Desk, and Bookshelf Status Are Too Small and Hard to Notice

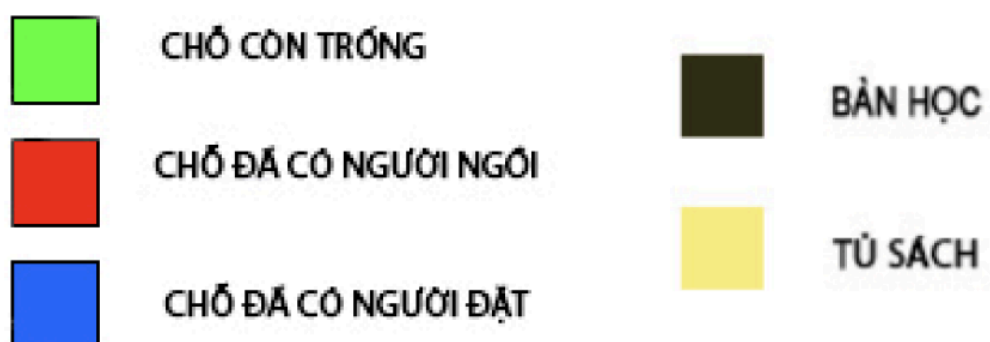


Figure 2. Legend for the Seating Diagram

In the current interface, the legends used to distinguish the status of seats, study desks, and bookshelves are displayed with very small text and icons, and are placed in an

inconspicuous area on the screen. This makes it difficult for users to quickly recognize essential information, forcing them to carefully search and examine the display to understand the meaning of each color or symbol.

According to the **visibility principle** in Human-Computer Interaction (HCI), all key elements that support user navigation and decision-making should be presented clearly, prominently, and understandably. However, with the current design, the legends are not visually prominent and do not effectively support quick recognition. This poses particular challenges for new users, the elderly, or users in low-light environments.

Additionally, the need to constantly cross-reference between the seating diagram and the small legend interrupts the user flow, slows down interaction, and increases the likelihood of errors — such as mistakenly selecting a reserved seat.

Lack of Intuitive Icons to Support Users

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Figure 3. Seat Number and Real-Time Availability Display in the Reservation Layout

The current interface mainly relies on text and color to convey information (e.g., seat status, area type, or specific functions of each section), but lacks accompanying intuitive icons. This forces users to spend time reading and interpreting the content, instead of quickly understanding it through familiar visual cues.

In HCI, the use of intuitive icons is a key principle for enhancing memorability, quick recognition, and reducing the cognitive load of reading. The absence of icons makes the interface text-heavy, less accessible, and less user-friendly — especially for new users, older adults, or those with limited technological experience. Furthermore, relying solely on color without accompanying visual symbols can pose difficulties for users with color blindness, making it hard for them to distinguish between different statuses.

Important Information Not Emphasized

Bạn có muốn xác nhận đặt
chỗ tại khu vực A09 không ?



Figure 4. Reservation Confirmation Popup

In the current confirmation popup, the displayed message reads:

"Do you want to confirm your reservation in area A09?"

However, the phrase **"area A09"** — which is the most crucial piece of information for the user's decision-making — is not visually highlighted. As a result, users may overlook it or need to reread the entire sentence to be sure.

To improve this, the team suggests using **bold formatting and/or contrasting colors** to emphasize important information. For example:

"Do you want to confirm your reservation in area A09?"

This approach enhances visual emphasis on critical information, supports faster recognition (recognition over recall), and improves decision accuracy.

Buttons Are Not Visually Prominent

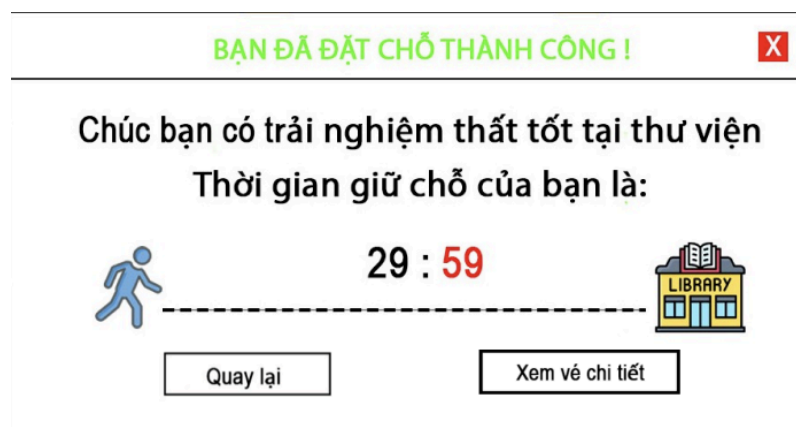


Figure 5. Successful Reservation Confirmation

After the user completes the reservation, the system displays a success message along with a countdown timer indicating the time left to hold the seat. However, this interface does not clearly guide the user on what to do next.

The two buttons — “*Go Back*” and “*View Ticket Details*” — are placed at the bottom of the screen but are not designed to be visually prominent. Both use a white background with thin black borders, which makes them appear dull and fail to attract attention.

Users may not know which action to take next, as both buttons have identical designs and lack visual emphasis, such as a primary color, larger size, or action-supporting icons.

The absence of clear guidance or visual hierarchy may confuse users, causing them to hesitate or even skip an important step like viewing their detailed ticket.

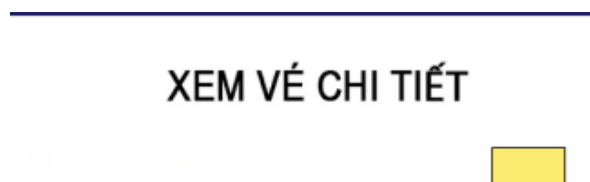


Figure 6. “View Ticket Details” Button

The current interface presents the “View Ticket Details” button with black text on a white background, without any border, icon, or standout effect. This minimalist design makes the button blend in with the overall layout of the screen.

Users are not provided with any visual cues (such as bold colors, hover effects, illustrative icons, or increased size) to indicate that this is an important action to take after a successful reservation.

Because it lacks visual prominence, the “View Ticket Details” button may be overlooked or not recognized as a clickable element. This undermines the user experience and may prevent users from accessing their booking information.

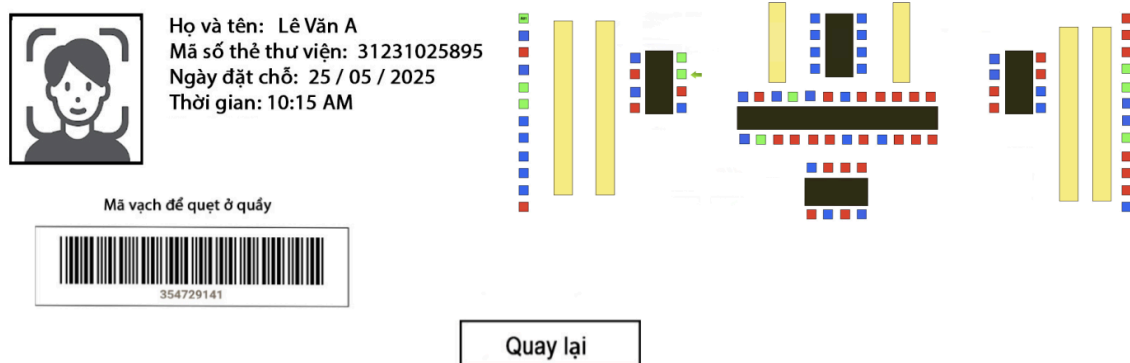


Figure 7. “View Ticket Details” Page

Unclear Functional Buttons

The spatial layout in the current interface is overly condensed, making details cluttered and difficult to see. Users struggle to clearly distinguish their reserved seat from other areas in the library. This negatively impacts their ability to orient themselves and navigate the physical space after booking.

Ticket and Seat Information Are Not Visually Intuitive

Although a green arrow is used to indicate the selected seat, this design is not intuitive or easy to understand. Using only a small green arrow in a densely marked diagram makes it easy for users to overlook or misinterpret their location. This could lead to confusion when trying to find the right area in the actual library.

The “Go Back” Button Still Lacks Prominence

As seen on previous screens, the “Go Back” button is again designed too simply — with a white background, thin black border, and no supporting icon or visual emphasis. This design lacks visual appeal and fails to clearly communicate its navigational function, especially after viewing the ticket.

2. Alternative Design Approaches Considered

During the course of this project, I personally considered several design alternatives that differed from the final version selected by the team.

One of my suggestions was to use a 3D space map interface or semi-isometric perspective, allowing users to visualize the room layout more intuitively — including walkways, entrances, and functional zones like bookshelves or group study areas. Unlike the 2D schematic chosen by the team — which feels more technical and

abstract — this approach could offer a more realistic simulation, thereby helping users with orientation and decision-making.

Another idea I explored was to implement a zone-based reservation model, where users select a general area such as “Quiet Zone” or “Power Outlet Area,” and the system automatically suggests a suitable seat. This method simplifies the reservation process and avoids overwhelming users with too much information from a detailed seat map. However, the team prioritized a more granular design, allowing users full control over seat selection, so this proposal was not adopted.

The key difference between these approaches lies in the level of detail and the decision-making support provided. The team’s design emphasizes user control and complete information, while my alternatives leaned toward reducing cognitive load and enhancing visual and spatial experience.

3. If Given the Opportunity to Redesign, a Different Prototype Would Be Created

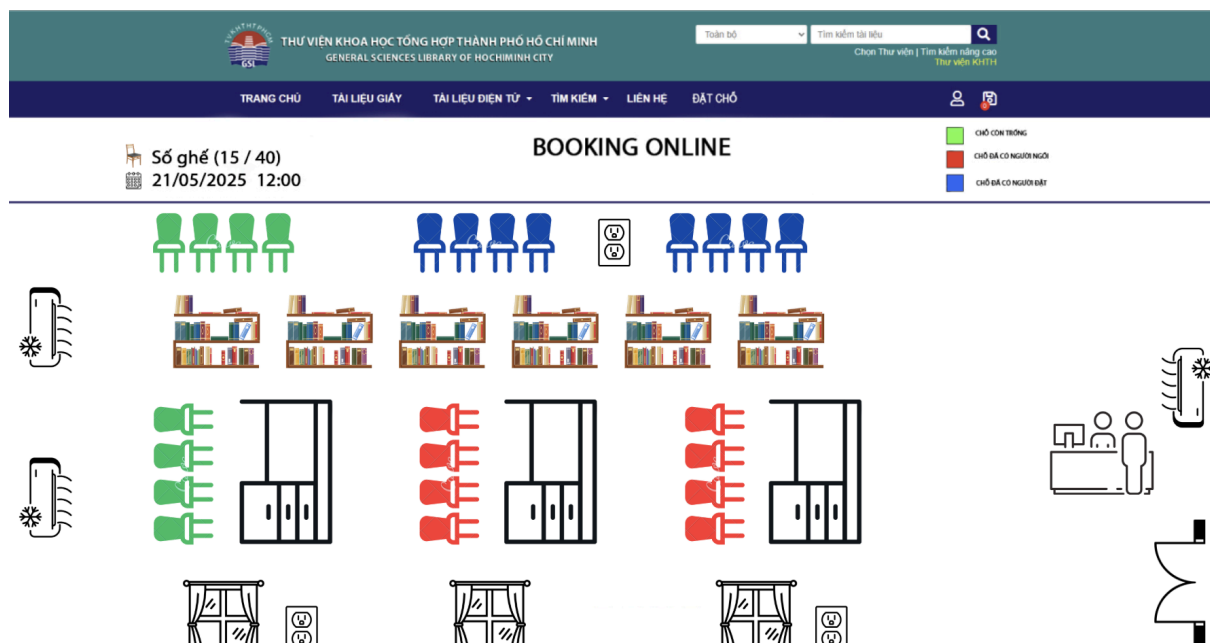


Figure 8. Customized Homepage of the Reservation Feature

Improved Seat Map Layout: I would redesign the seat map to make it clearer and more interactive. Additionally, I would incorporate a **seat filtering function** based on user preferences (e.g., near windows, power outlets, air conditioning...) to help users quickly find a suitable spot.

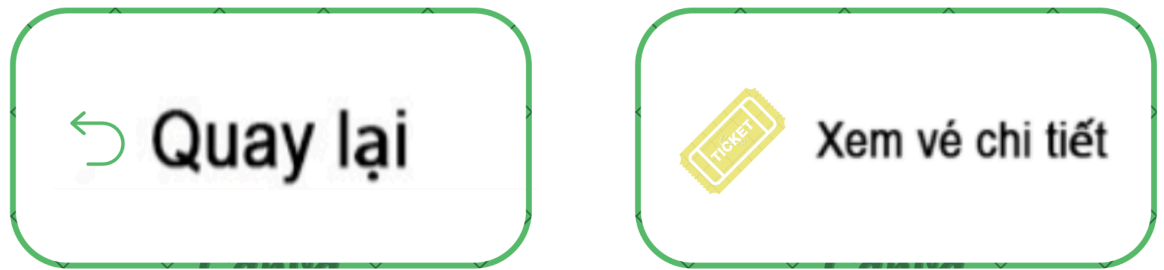


Figure 9. Redesigned “Back” and “View Ticket Details” Buttons

Improved Functional Buttons: Buttons like “Back” and “View Ticket Details” would be made more prominent by using appropriate color schemes, adding illustrative icons, and modifying their size and border styles to create stronger visual emphasis.

Enhanced Visual Cues on the Ticket Details Page

Currently, users find it difficult to identify the seat location after booking. Therefore, I would implement the following improvements:

- **Highlight the selected seat** on the map more clearly (e.g., using bold outlines or distinct colors), along with a short description below such as: *“You selected seat A09 – quiet zone near the window.”*
- **Integrate real-world images** or a simple **3D rendering** of the space to help users visualize the actual environment more effectively.

Add a Seat Cancellation Feature

One notable shortcoming in the current prototype is the absence of a function to cancel booked seats. In reality, users may change plans, book incorrectly, or no longer need the reserved spot. If the system doesn’t allow cancellations, users may feel a lack of control, resulting in a negative and rigid experience. Moreover, reserved but unused seats continue to occupy space in the system, wasting library resources.

Therefore, I propose adding a “Cancel Reservation” feature to the interface, especially on the “View Ticket Details” page or in a booking management section. The cancel button should be clearly visible, easily accessible, and placed in a convenient location. To prevent accidental actions, the system should also display a confirmation dialog, such as:

“Are you sure you want to cancel the reserved seat in area A09?”

This design not only follows key HCI principles like “User Control and Freedom” and “Error Prevention”, but also enhances overall user satisfaction, ensuring the system is more efficient and flexible in real-world usage.

4. What do you think about the user evaluation process? If you could do things differently, what would you change?

Expanding the Scale and Diversity of Participants

Expanding both the scale and diversity of participant groups is crucial to obtaining a more comprehensive and objective view of the user experience. Testing with only a small or homogeneous group might lead to missing out on issues or unique needs from different types of users.

A **larger number of participants** helps uncover more usability issues. Studies have shown that usability problems tend to repeat and reach saturation after a certain number of users. However, with too few users (like just 5), there's still a high chance that you haven't reached that saturation point, potentially missing out on important errors or confusing elements.

Increasing diversity: Although the main target audience was second-year students, including participants with different needs or levels of experience would enrich the feedback. For example:

- First-year students
- Final-year or postgraduate students
- Faculty members or library staff (if applicable)

Emphasizing diversity in user testing reflects thoughtful UX practice — it shows that the evaluation isn't just about functionality, but about building a more inclusive and comprehensive user experience.

Considering the Think-Aloud Protocol

The “think-aloud” method is a powerful tool that gives you deeper insight into the user's cognitive process while interacting with the product. Rather than only observing external actions, you also get to “hear” their internal reasoning.

How it works: During the task, ask users to verbalize their thoughts — what they're doing, seeing, expecting, or questioning at each moment.

Example: “I'm looking for the booking button... oh, there it is... but why is it so faded? I'm not sure if I should click it...”

Digging into the “why”: This method helps uncover the reasons behind user behavior. If someone clicks the wrong button but says, “I thought this meant cancel,” it reveals confusion about icons or layout. It helps you discover *invisible friction* —

where users don't make obvious errors but struggle internally to understand something.

Uncovering latent needs: Sometimes users will say things like, “I wish there was a...” or “If only I could do A instead of B...” These spontaneous remarks are clues to unspoken needs or features they desire but can’t articulate through surveys.

Proposing this method shows that you’re not just focused on task completion, but also on deeply understanding the *internal experience* of the user — leading to more strategic improvements.

Testing on More Devices

In today’s multi-device world, it’s not enough for a website to work well on a desktop. Users expect seamless experiences across all devices they use.

Even though your team implemented responsive design, actual testing on various devices brings several benefits:

- **Evaluate physical interaction:** Using a phone involves touch, not clicks. Are buttons large enough to tap easily? Are elements spaced out to prevent accidental touches?
- **Catch display issues:** Problems like overflow, layout bugs, or images not rendering correctly may only show up on specific screen sizes or browsers.
- **Mobile performance matters:** Load times and animation smoothness can vary between desktops and phones, especially on slower mobile networks.
- **Relevance to students:** For students, booking library space via mobile or tablet while on the go is common. Poor mobile experiences may deter them from using the system at all.

Highlighting the importance of multi-device testing demonstrates a practical, up-to-date UX mindset — recognizing that good design means adaptability and accessibility across platforms.

Testing “Worst-Case Scenarios”

A strong product doesn't just work smoothly in perfect conditions — it needs to be robust when things go wrong. Worst-case scenario testing helps evaluate how the system handles errors and communicates with users.

Purpose: To assess how both the system and the users react under less-than-ideal circumstances or system failures.

Common scenarios to test:

- **Failed booking:** Someone tries to reserve a seat that's already taken, or the system encounters an error. Does it give a clear error message? Suggest alternatives?
- **Incorrect data entry:** The user inputs an invalid date or personal info. Does the system point out the mistake and guide the user?
- **Loss of connection:** If the internet drops mid-task, is progress saved? Can the user resume later?
- **Illogical choices:** What happens if the user selects a time outside library hours or a past date? Are there warnings or restrictions?

Benefit: Ensures frictionless recovery. When users face errors, they need clear messages, explanations, and instructions — not confusion or dead ends. This builds trust and loyalty, as users appreciate systems that help them recover gracefully.

By proposing this kind of testing, you show that UX isn't just about good looks or features, but also about resilience and stability in real-world use.

User Journey Mapping

While task-based testing helps evaluate specific functions, user journey mapping provides a holistic view of the entire user experience with the system.

Goal: To understand the full “story” the user goes through — from the moment they feel a need, to when they complete their goal and reflect on the experience.

How to do it:

Create a visual map that outlines the steps the user takes, and note:

- **Stages:** Key phases of the journey (e.g., searching for seats, booking, managing bookings).
- **Touchpoints:** Where the user interacts with the system (homepage, seat map, confirmation form, email, chatbot).
- **Actions:** What the user does at each touchpoint.
- **Emotions:** How the user feels (e.g., excited, confused, frustrated, satisfied). You can gather this from think-aloud comments or emotional cues from videos.
- **Pain points:** Where users struggle or feel discomfort.

- **Opportunities:** Ideas to make the journey smoother, more enjoyable, or more efficient.

Benefits:

- **Comprehensive view:** Helps you spot cross-functional or transition issues that single task tests may miss.
- **Set priorities:** Once you see the full picture, it's easier to decide which pain points matter most and should be fixed first.
- **Effective communication:** A journey map is a visual, compelling way to present UX findings and ideas to stakeholders like developers or product managers.

Proposing this method shows that you're not just analyzing in fragments — you're connecting everything into a clear, strategic narrative that supports meaningful design improvements.